Cargo robot v2 - all connections

Graduation Project II - 22 October 2026. Every wire of the robot: the power wires (P), the signal wires (S), the small parts on the I/O board, then the connections of each part. The 4 schematic sheets follow at the end (A3). GPIO numbers come from firmware/cargo_robot/config.h.

How to read it. One row = one wire, from the first place to the second. Rows with the same ID belong to one job (e.g. S6 = the VCC of both motor drivers). **Label every wire with its ID at both ends.** Buses: WAGO +12 V MAIN (after the switch), WAGO +12 V MOTOR (after the E-stop), WAGO +5 V, WAGO +7.4 V and WAGO GND are lever connectors on the bottom plate; the 3V3 bus and GND bus are rows of linked pins on the I/O board (3V3 comes from the ESP32's 3V3 pin).

Never connect 5 V or more to an ESP32 pin, the MCP23017, the encoders, the HX711s or the line sensor: they all run on 3.3 V. Wire with the master switch OFF and the battery unplugged.

Power wires

ID	From	To	Wire	Note
P1	Battery BT1 **+ (XT60)**	Fuse holder F1 **in**	16 AWG red, 10 cm	XT60 soldered
P2	Fuse holder F1 **out**	Master switch SW1 **terminal 1**	16 AWG red	
P3	Master switch SW1 **terminal 2**	WAGO +12 V MAIN	16 AWG red	WAGO 221-415
P4	Battery BT1 **- (XT60)**	WAGO GND	16 AWG black	WAGO 221-415, link a second one
P5	WAGO +12 V MAIN	E-stop S1 **NC terminal 1**	16 AWG red, 40 cm	ferrule
P6	E-stop S1 **NC terminal 2**	WAGO +12 V MOTOR	16 AWG red, 40 cm	ferrule
P7	WAGO +12 V MOTOR	BTS7960 left U2 **B+**	16 AWG red	ferrule
P8	WAGO +12 V MOTOR	BTS7960 right U3 **B+**	16 AWG red	ferrule
P9	WAGO GND	BTS7960 left U2 **B-**	16 AWG black	ferrule
P10	WAGO GND	BTS7960 right U3 **B-**	16 AWG black	ferrule
P11	BTS7960 left U2 **M+**	Left motor M1 **red (motor +)**	20 AWG or the motor's wire	ferrule
P11	BTS7960 left U2 **M-**	Left motor M1 **white (motor -)**	20 AWG or the motor's wire	ferrule
P12	BTS7960 right U3 **M+**	Right motor M2 **red (motor +)**	20 AWG or the motor's wire	ferrule
P12	BTS7960 right U3 **M-**	Right motor M2 **white (motor -)**	20 AWG or the motor's wire	ferrule
P13	WAGO +12 V MAIN	5 V buck U1 **IN+**	20 AWG red	ferrule
P14	WAGO GND	5 V buck U1 **IN-**	20 AWG black	ferrule
P15	5 V buck U1 **OUT+**	WAGO +5 V	20 AWG red	set to 5.10 V first
P15	5 V buck U1 **OUT-**	WAGO GND	20 AWG black	
P15	WAGO +5 V	ESP32 (breakout) **5V**	20 AWG red	
P15	WAGO GND	ESP32 (breakout) **GND**	20 AWG black	the logic ground starts here

ID	From	To	Wire	Note
P16	Battery BT1 **balance plug**	LiPo voltage alarm	-	set it to 3.5 V per cell
P17	WAGO +12 V MOTOR	Servo buck U5 **IN+**	16 AWG red	ferrule
P18	WAGO GND	Servo buck U5 **IN-**	16 AWG black	ferrule
P19	Servo buck U5 **OUT+**	WAGO +7.4 V	16 AWG red	set to 7.4 V first
P19	Servo buck U5 **OUT-**	WAGO GND	16 AWG black	
P19	WAGO +7.4 V	Front servo SV1 **red (+)**	servo lead	
P19	WAGO GND	Front servo SV1 **brown (-)**	servo lead	
P19	WAGO +7.4 V	Rear servo SV2 **red (+)**	servo lead	
P19	WAGO GND	Rear servo SV2 **brown (-)**	servo lead	

Signal wires

ID	From	To	Wire	Note
S1	BTS7960 left U2 **RPWM**	ESP32 (breakout) **GPIO 25**	jumper 24-26 AWG	forward PWM
S2	BTS7960 left U2 **LPWM**	ESP32 (breakout) **GPIO 26**	jumper 24-26 AWG	reverse PWM
S3	BTS7960 right U3 **RPWM**	ESP32 (breakout) **GPIO 27**	jumper 24-26 AWG	forward PWM
S4	BTS7960 right U3 **LPWM**	ESP32 (breakout) **GPIO 14**	jumper 24-26 AWG	reverse PWM
S5	BTS7960 left U2 **R_EN + L_EN (linked)**	MCP23017 U4 **GPB0**	jumper 24-26 AWG	MOTOR_EN; 10k pull-down to GND
S5	BTS7960 right U3 **R_EN + L_EN (linked)**	MCP23017 U4 **GPB0**	jumper 24-26 AWG	MOTOR_EN
S6	BTS7960 left U2 **VCC**	3V3 bus	jumper 24-26 AWG	3.3 V, not 5 V
S6	BTS7960 right U3 **VCC**	3V3 bus	jumper 24-26 AWG	3.3 V, not 5 V
S7	BTS7960 left U2 **GND (logic)**	GND bus	jumper 24-26 AWG	
S7	BTS7960 right U3 **GND (logic)**	GND bus	jumper 24-26 AWG	
S8	Left motor M1 **blue (encoder VCC)**	3V3 bus	jumper 24-26 AWG	3.3 V, never 5 V
S9	Left motor M1 **black (encoder GND)**	GND bus	jumper 24-26 AWG	
S10	Left motor M1 **yellow (encoder A)**	ESP32 (breakout) **GPIO 34**	jumper 24-26 AWG	10k pull-up to 3V3
S11	Left motor M1 **green (encoder B)**	ESP32 (breakout) **GPIO 35**	jumper 24-26 AWG	10k pull-up to 3V3
S12	Right motor M2 **blue (encoder VCC)**	3V3 bus	jumper 24-26 AWG	3.3 V, never 5 V
S13	Right motor M2 **black (encoder GND)**	GND bus	jumper 24-26 AWG	
S14	Right motor M2 **yellow (encoder A)**	ESP32 (breakout) **GPIO 36 (VP)**	jumper 24-26 AWG	10k pull-up to 3V3
S15	Right motor M2 **green (encoder B)**	ESP32 (breakout) **GPIO 39 (VN)**	jumper 24-26 AWG	10k pull-up to 3V3
S16	HX711 front-left **VCC**	3V3 bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S16	HX711 front-right **VCC**	3V3 bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S16	HX711 rear-left **VCC**	3V3 bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S16	HX711 rear-right **VCC**	3V3 bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17	HX711 front-left **GND**	GND bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17	HX711 front-right **GND**	GND bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17	HX711 rear-left **GND**	GND bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17	HX711 rear-right **GND**	GND bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	HX711 front-left **SCK**	ESP32 (breakout) **GPIO 5**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	HX711 front-right **SCK**	ESP32 (breakout) **GPIO 5**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	HX711 rear-left **SCK**	ESP32 (breakout) **GPIO 5**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	HX711 rear-right **SCK**	ESP32 (breakout) **GPIO 5**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S19	HX711 front-left **DT**	ESP32 (breakout) **GPIO 16**	jumper 24-26 AWG	
S20	HX711 front-right **DT**	ESP32 (breakout) **GPIO 17**	jumper 24-26 AWG	
S21	HX711 rear-left **DT**	ESP32 (breakout) **GPIO 18**	jumper 24-26 AWG	
S22	HX711 rear-right **DT**	ESP32 (breakout) **GPIO 19**	jumper 24-26 AWG	

ID	From	To	Wire	Note
S23	Load cell front-left **red**	HX711 front-left **E+**	the cell's own wires	
S23	Load cell front-left **black**	HX711 front-left **E-**	the cell's own wires	
S23	Load cell front-left **white**	HX711 front-left **A-**	the cell's own wires	
S23	Load cell front-left **green**	HX711 front-left **A+**	the cell's own wires	
S23	Load cell front-right **red**	HX711 front-right **E+**	the cell's own wires	
S23	Load cell front-right **black**	HX711 front-right **E-**	the cell's own wires	
S23	Load cell front-right **white**	HX711 front-right **A-**	the cell's own wires	
S23	Load cell front-right **green**	HX711 front-right **A+**	the cell's own wires	
S23	Load cell rear-left **red**	HX711 rear-left **E+**	the cell's own wires	
S23	Load cell rear-left **black**	HX711 rear-left **E-**	the cell's own wires	
S23	Load cell rear-left **white**	HX711 rear-left **A-**	the cell's own wires	
S23	Load cell rear-left **green**	HX711 rear-left **A+**	the cell's own wires	
S23	Load cell rear-right **red**	HX711 rear-right **E+**	the cell's own wires	
S23	Load cell rear-right **black**	HX711 rear-right **E-**	the cell's own wires	
S23	Load cell rear-right **white**	HX711 rear-right **A-**	the cell's own wires	
S23	Load cell rear-right **green**	HX711 rear-right **A+**	the cell's own wires	
S24	HC-SR04 (front) **VCC**	WAGO +5 V	jumper 24-26 AWG	5 V
S25	HC-SR04 (front) **GND**	GND bus	jumper 24-26 AWG	
S26	HC-SR04 (front) **TRIG**	ESP32 (breakout) **GPIO 4**	jumper 24-26 AWG	
S27	HC-SR04 (front) **ECHO**	ESP32 (breakout) **GPIO 23**	jumper 24-26 AWG	through a 1k / 2k divider (5 V -> 3.3 V)
S28	Front servo SV1 **orange (signal)**	ESP32 (breakout) **GPIO 2**	servo lead	
S29	Rear servo SV2 **orange (signal)**	ESP32 (breakout) **GPIO 12**	servo lead	nothing else on this pin (must be LOW at start-up)
S30	ESP32 (breakout) **GPIO 21 (SDA)**	MCP23017 U4 **SDA**	jumper 24-26 AWG	4.7k pull-up to 3V3
S30	ESP32 (breakout) **GPIO 21 (SDA)**	VL53L0X **SDA**	jumper 24-26 AWG	
S31	ESP32 (breakout) **GPIO 22 (SCL)**	MCP23017 U4 **SCL**	jumper 24-26 AWG	4.7k pull-up to 3V3
S31	ESP32 (breakout) **GPIO 22 (SCL)**	VL53L0X **SCL**	jumper 24-26 AWG	
S32	ESP32 (breakout) **3V3**	3V3 bus	jumper 24-26 AWG	feeds every 3.3 V part
S32	ESP32 (breakout) **GND**	GND bus	jumper 24-26 AWG	
S32	MCP23017 U4 **VDD, RESET**	3V3 bus	jumper 24-26 AWG	
S32	MCP23017 U4 **VSS, A0, A1, A2**	GND bus	jumper 24-26 AWG	address 0x20
S32	VL53L0X **VIN**	3V3 bus	jumper 24-26 AWG	
S32	VL53L0X **GND**	GND bus	jumper 24-26 AWG	
S33	Line sensor **VCC**	3V3 bus	jumper 24-26 AWG	
S33	Line sensor **GND**	GND bus	jumper 24-26 AWG	

ID	From	To	Wire	Note
S34	Line sensor **OUT1**	MCP23017 U4 **GPA0**	jumper 24-26 AWG	OUT1 = leftmost
S34	Line sensor **OUT2**	MCP23017 U4 **GPA1**	jumper 24-26 AWG	
S34	Line sensor **OUT3**	MCP23017 U4 **GPA2**	jumper 24-26 AWG	
S34	Line sensor **OUT4**	MCP23017 U4 **GPA3**	jumper 24-26 AWG	
S34	Line sensor **OUT5**	MCP23017 U4 **GPA4**	jumper 24-26 AWG	
S34	Line sensor **OUT6**	MCP23017 U4 **GPA5**	jumper 24-26 AWG	
S34	Line sensor **OUT7**	MCP23017 U4 **GPA6**	jumper 24-26 AWG	
S34	Line sensor **OUT8**	MCP23017 U4 **GPA7**	jumper 24-26 AWG	OUT8 = rightmost
S35	Start button S2 **terminal 1**	MCP23017 U4 **GPB4**	jumper 24-26 AWG	internal pull-up
S35	Start button S2 **terminal 2**	GND bus	jumper 24-26 AWG	
S36	Green LED D1 **+ (long leg)**	MCP23017 U4 **GPB2**	jumper 24-26 AWG	through 330 ohm
S36	Green LED D1 **- (short leg)**	GND bus	jumper 24-26 AWG	
S36	Red LED D2 **+ (long leg)**	MCP23017 U4 **GPB3**	jumper 24-26 AWG	through 330 ohm
S36	Red LED D2 **- (short leg)**	GND bus	jumper 24-26 AWG	
S37	Buzzer module **VCC**	3V3 bus	jumper 24-26 AWG	
S37	Buzzer module **GND**	GND bus	jumper 24-26 AWG	
S37	Buzzer module **I/O**	MCP23017 U4 **GPB1**	jumper 24-26 AWG	
S38	Bus-sense divider **top of 10k**	WAGO +12 V MOTOR	jumper 24-26 AWG	after the E-stop
S38	Bus-sense divider **bottom of 2.2k**	GND bus	jumper 24-26 AWG	100 nF across the 2.2k
S38	Bus-sense divider **junction**	ESP32 (breakout) **GPIO 33**	jumper 24-26 AWG	about 2.3 V at 12.6 V; 0 V = E-stop pressed

The I/O board (5 x 7 cm perfboard, schematic E04)

Part	Connected
MCP23017 module	I2C I/O expander, address 0x20 (A0-A2 to GND, RESET to 3V3)
2 x 4.7k	SDA -> 3V3, SCL -> 3V3
1 x 10k	MOTOR_EN (GPB0) -> GND: motors off until the ESP32 has started
4 x 10k	encoder lines GPIO 34, 35, 36, 39 -> 3V3 (these pins have no pull-ups inside)
1k + 2k	ultrasonic ECHO divider: ECHO - 1k - junction - 2k - GND; junction to the ESP32
10k + 2.2k + 100 nF	bus sense: +12 V MOTOR - 10k - junction - 2.2k - GND; junction to GPIO 33
2 x 330	in series with the green and red LEDs
3 rows of linked pins	3V3 bus, GND bus, 5 V (from WAGO +5 V)

Connections of each part

The same wires again, listed under every part they touch: use it to check one part at a time.

Battery BT1

ID	Pin here	Goes to	Wire	Note
P1	+ (XT60)	Fuse holder F1 **in**	16 AWG red, 10 cm	XT60 soldered
P4	- (XT60)	WAGO GND	16 AWG black	WAGO 221-415, link a second one
P16	balance plug	LiPo voltage alarm	-	set it to 3.5 V per cell

Fuse holder F1

ID	Pin here	Goes to	Wire	Note
P1	in	Battery BT1 **+ (XT60)**	16 AWG red, 10 cm	XT60 soldered
P2	out	Master switch SW1 **terminal 1**	16 AWG red	

Master switch SW1

ID	Pin here	Goes to	Wire	Note
P2	terminal 1	Fuse holder F1 **out**	16 AWG red	
P3	terminal 2	WAGO +12 V MAIN	16 AWG red	WAGO 221-415

WAGO +12 V MAIN

ID	Pin here	Goes to	Wire	Note
P3		Master switch SW1 **terminal 2**	16 AWG red	WAGO 221-415
P5		E-stop S1 **NC terminal 1**	16 AWG red, 40 cm	ferrule
P13		5 V buck U1 **IN+**	20 AWG red	ferrule

WAGO GND

ID	Pin here	Goes to	Wire	Note
P4		Battery BT1 **- (XT60)**	16 AWG black	WAGO 221-415, link a second one
P9		BTS7960 left U2 **B-**	16 AWG black	ferrule
P10		BTS7960 right U3 **B-**	16 AWG black	ferrule
P14		5 V buck U1 **IN-**	20 AWG black	ferrule
P15		5 V buck U1 **OUT-**	20 AWG black	
P15		ESP32 (breakout) **GND**	20 AWG black	the logic ground starts here
P18		Servo buck U5 **IN-**	16 AWG black	ferrule
P19		Servo buck U5 **OUT-**	16 AWG black	
P19		Front servo SV1 **brown (-)**	servo lead	
P19		Rear servo SV2 **brown (-)**	servo lead	

E-stop S1

ID	Pin here	Goes to	Wire	Note
P5	NC terminal 1	WAGO +12 V MAIN	16 AWG red, 40 cm	ferrule
P6	NC terminal 2	WAGO +12 V MOTOR	16 AWG red, 40 cm	ferrule

WAGO +12 V MOTOR

ID	Pin here	Goes to	Wire	Note
P6		E-stop S1 **NC terminal 2**	16 AWG red, 40 cm	ferrule
P7		BTS7960 left U2 **B+**	16 AWG red	ferrule
P8		BTS7960 right U3 **B+**	16 AWG red	ferrule
P17		Servo buck U5 **IN+**	16 AWG red	ferrule
S38		Bus-sense divider **top of 10k**	jumper 24-26 AWG	after the E-stop

BTS7960 left U2

ID	Pin here	Goes to	Wire	Note
P7	B+	WAGO +12 V MOTOR	16 AWG red	ferrule
P9	B-	WAGO GND	16 AWG black	ferrule
P11	M+	Left motor M1 **red (motor +)**	20 AWG or the motor's wire	ferrule
P11	M-	Left motor M1 **white (motor -)**	20 AWG or the motor's wire	ferrule
S1	RPWM	ESP32 (breakout) **GPIO 25**	jumper 24-26 AWG	forward PWM
S2	LPWM	ESP32 (breakout) **GPIO 26**	jumper 24-26 AWG	reverse PWM
S5	R_EN + L_EN (linked)	MCP23017 U4 **GPB0**	jumper 24-26 AWG	MOTOR_EN; 10k pull-down to GND
S6	VCC	3V3 bus	jumper 24-26 AWG	3.3 V, not 5 V
S7	GND (logic)	GND bus	jumper 24-26 AWG	

BTS7960 right U3

ID	Pin here	Goes to	Wire	Note
P8	B+	WAGO +12 V MOTOR	16 AWG red	ferrule
P10	B-	WAGO GND	16 AWG black	ferrule
P12	M+	Right motor M2 **red (motor +)**	20 AWG or the motor's wire	ferrule
P12	M-	Right motor M2 **white (motor -)**	20 AWG or the motor's wire	ferrule
S3	RPWM	ESP32 (breakout) **GPIO 27**	jumper 24-26 AWG	forward PWM
S4	LPWM	ESP32 (breakout) **GPIO 14**	jumper 24-26 AWG	reverse PWM
S5	R_EN + L_EN (linked)	MCP23017 U4 **GPB0**	jumper 24-26 AWG	MOTOR_EN
S6	VCC	3V3 bus	jumper 24-26 AWG	3.3 V, not 5 V
S7	GND (logic)	GND bus	jumper 24-26 AWG	

Left motor M1

ID	Pin here	Goes to	Wire	Note
P11	red (motor +)	BTS7960 left U2 **M+**	20 AWG or the motor's wire	ferrule
P11	white (motor -)	BTS7960 left U2 **M-**	20 AWG or the motor's wire	ferrule
S8	blue (encoder VCC)	3V3 bus	jumper 24-26 AWG	3.3 V, never 5 V
S9	black (encoder GND)	GND bus	jumper 24-26 AWG	
S10	yellow (encoder A)	ESP32 (breakout) **GPIO 34**	jumper 24-26 AWG	10k pull-up to 3V3
S11	green (encoder B)	ESP32 (breakout) **GPIO 35**	jumper 24-26 AWG	10k pull-up to 3V3

Right motor M2

ID	Pin here	Goes to	Wire	Note
P12	red (motor +)	BTS7960 right U3 **M+**	20 AWG or the motor's wire	ferrule
P12	white (motor -)	BTS7960 right U3 **M-**	20 AWG or the motor's wire	ferrule
S12	blue (encoder VCC)	3V3 bus	jumper 24-26 AWG	3.3 V, never 5 V
S13	black (encoder GND)	GND bus	jumper 24-26 AWG	
S14	yellow (encoder A)	ESP32 (breakout) **GPIO 36 (VP)**	jumper 24-26 AWG	10k pull-up to 3V3
S15	green (encoder B)	ESP32 (breakout) **GPIO 39 (VN)**	jumper 24-26 AWG	10k pull-up to 3V3

5 V buck U1

ID	Pin here	Goes to	Wire	Note
P13	IN+	WAGO +12 V MAIN	20 AWG red	ferrule
P14	IN-	WAGO GND	20 AWG black	ferrule
P15	OUT+	WAGO +5 V	20 AWG red	set to 5.10 V first
P15	OUT-	WAGO GND	20 AWG black	

WAGO +5 V

ID	Pin here	Goes to	Wire	Note
P15		5 V buck U1 **OUT+**	20 AWG red	set to 5.10 V first
P15		ESP32 (breakout) **5V**	20 AWG red	
S24		HC-SR04 (front) **VCC**	jumper 24-26 AWG	5 V

ESP32 (breakout)

ID	Pin here	Goes to	Wire	Note
P15	5V	WAGO +5 V	20 AWG red	
P15	GND	WAGO GND	20 AWG black	the logic ground starts here
S1	GPIO 25	BTS7960 left U2 **RPWM**	jumper 24-26 AWG	forward PWM
S2	GPIO 26	BTS7960 left U2 **LPWM**	jumper 24-26 AWG	reverse PWM
S3	GPIO 27	BTS7960 right U3 **RPWM**	jumper 24-26 AWG	forward PWM
S4	GPIO 14	BTS7960 right U3 **LPWM**	jumper 24-26 AWG	reverse PWM
S10	GPIO 34	Left motor M1 **yellow (encoder A)**	jumper 24-26 AWG	10k pull-up to 3V3
S11	GPIO 35	Left motor M1 **green (encoder B)**	jumper 24-26 AWG	10k pull-up to 3V3
S14	GPIO 36 (VP)	Right motor M2 **yellow (encoder A)**	jumper 24-26 AWG	10k pull-up to 3V3
S15	GPIO 39 (VN)	Right motor M2 **green (encoder B)**	jumper 24-26 AWG	10k pull-up to 3V3
S18	GPIO 5	HX711 front-left **SCK**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	GPIO 5	HX711 front-right **SCK**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	GPIO 5	HX711 rear-left **SCK**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	GPIO 5	HX711 rear-right **SCK**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S19	GPIO 16	HX711 front-left **DT**	jumper 24-26 AWG	
S20	GPIO 17	HX711 front-right **DT**	jumper 24-26 AWG	
S21	GPIO 18	HX711 rear-left **DT**	jumper 24-26 AWG	
S22	GPIO 19	HX711 rear-right **DT**	jumper 24-26 AWG	
S26	GPIO 4	HC-SR04 (front) **TRIG**	jumper 24-26 AWG	
S27	GPIO 23	HC-SR04 (front) **ECHO**	jumper 24-26 AWG	through a 1k / 2k divider (5 V -> 3.3 V)
S28	GPIO 2	Front servo SV1 **orange (signal)**	servo lead	
S29	GPIO 12	Rear servo SV2 **orange (signal)**	servo lead	nothing else on this pin (must be LOW at start-up)
S30	GPIO 21 (SDA)	MCP23017 U4 **SDA**	jumper 24-26 AWG	4.7k pull-up to 3V3
S30	GPIO 21 (SDA)	VL53L0X **SDA**	jumper 24-26 AWG	
S31	GPIO 22 (SCL)	MCP23017 U4 **SCL**	jumper 24-26 AWG	4.7k pull-up to 3V3
S31	GPIO 22 (SCL)	VL53L0X **SCL**	jumper 24-26 AWG	
S32	3V3	3V3 bus	jumper 24-26 AWG	feeds every 3.3 V part
S32	GND	GND bus	jumper 24-26 AWG	
S38	GPIO 33	Bus-sense divider **junction**	jumper 24-26 AWG	about 2.3 V at 12.6 V; 0 V = E-stop pressed

LiPo voltage alarm

ID	Pin here	Goes to	Wire	Note
P16		Battery BT1 **balance plug**	-	set it to 3.5 V per cell

Servo buck U5

ID	Pin here	Goes to	Wire	Note
P17	IN+	WAGO +12 V MOTOR	16 AWG red	ferrule
P18	IN-	WAGO GND	16 AWG black	ferrule
P19	OUT+	WAGO +7.4 V	16 AWG red	set to 7.4 V first
P19	OUT-	WAGO GND	16 AWG black	

WAGO +7.4 V

ID	Pin here	Goes to	Wire	Note
P19		Servo buck U5 **OUT+**	16 AWG red	set to 7.4 V first
P19		Front servo SV1 **red (+)**	servo lead	
P19		Rear servo SV2 **red (+)**	servo lead	

Front servo SV1

ID	Pin here	Goes to	Wire	Note
P19	red (+)	WAGO +7.4 V	servo lead	
P19	brown (-)	WAGO GND	servo lead	
S28	orange (signal)	ESP32 (breakout) **GPIO 2**	servo lead	

Rear servo SV2

ID	Pin here	Goes to	Wire	Note
P19	red (+)	WAGO +7.4 V	servo lead	
P19	brown (-)	WAGO GND	servo lead	
S29	orange (signal)	ESP32 (breakout) **GPIO 12**	servo lead	nothing else on this pin (must be LOW at start-up)

MCP23017 U4

ID	Pin here	Goes to	Wire	Note
S5	GPB0	BTS7960 left U2 **R_EN + L_EN (linked)**	jumper 24-26 AWG	MOTOR_EN; 10k pull-down to GND
S5	GPB0	BTS7960 right U3 **R_EN + L_EN (linked)**	jumper 24-26 AWG	MOTOR_EN
S30	SDA	ESP32 (breakout) **GPIO 21 (SDA)**	jumper 24-26 AWG	4.7k pull-up to 3V3
S31	SCL	ESP32 (breakout) **GPIO 22 (SCL)**	jumper 24-26 AWG	4.7k pull-up to 3V3
S32	VDD, RESET	3V3 bus	jumper 24-26 AWG	
S32	VSS, A0, A1, A2	GND bus	jumper 24-26 AWG	address 0x20
S34	GPA0	Line sensor **OUT1**	jumper 24-26 AWG	OUT1 = leftmost
S34	GPA1	Line sensor **OUT2**	jumper 24-26 AWG	
S34	GPA2	Line sensor **OUT3**	jumper 24-26 AWG	
S34	GPA3	Line sensor **OUT4**	jumper 24-26 AWG	
S34	GPA4	Line sensor **OUT5**	jumper 24-26 AWG	
S34	GPA5	Line sensor **OUT6**	jumper 24-26 AWG	
S34	GPA6	Line sensor **OUT7**	jumper 24-26 AWG	
S34	GPA7	Line sensor **OUT8**	jumper 24-26 AWG	OUT8 = rightmost
S35	GPB4	Start button S2 **terminal 1**	jumper 24-26 AWG	internal pull-up
S36	GPB2	Green LED D1 **+ (long leg)**	jumper 24-26 AWG	through 330 ohm
S36	GPB3	Red LED D2 **+ (long leg)**	jumper 24-26 AWG	through 330 ohm
S37	GPB1	Buzzer module **I/O**	jumper 24-26 AWG	

3V3 bus

ID	Pin here	Goes to	Wire	Note
S6		BTS7960 left U2 **VCC**	jumper 24-26 AWG	3.3 V, not 5 V
S6		BTS7960 right U3 **VCC**	jumper 24-26 AWG	3.3 V, not 5 V
S8		Left motor M1 **blue (encoder VCC)**	jumper 24-26 AWG	3.3 V, never 5 V
S12		Right motor M2 **blue (encoder VCC)**	jumper 24-26 AWG	3.3 V, never 5 V
S16		HX711 front-left **VCC**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S16		HX711 front-right **VCC**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S16		HX711 rear-left **VCC**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S16		HX711 rear-right **VCC**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S32		ESP32 (breakout) **3V3**	jumper 24-26 AWG	feeds every 3.3 V part
S32		MCP23017 U4 **VDD, RESET**	jumper 24-26 AWG	
S32		VL53L0X **VIN**	jumper 24-26 AWG	
S33		Line sensor **VCC**	jumper 24-26 AWG	
S37		Buzzer module **VCC**	jumper 24-26 AWG	

GND bus

ID	Pin here	Goes to	Wire	Note
S7		BTS7960 left U2 **GND (logic)**	jumper 24-26 AWG	
S7		BTS7960 right U3 **GND (logic)**	jumper 24-26 AWG	
S9		Left motor M1 **black (encoder GND)**	jumper 24-26 AWG	
S13		Right motor M2 **black (encoder GND)**	jumper 24-26 AWG	
S17		HX711 front-left **GND**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17		HX711 front-right **GND**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17		HX711 rear-left **GND**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17		HX711 rear-right **GND**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S25		HC-SR04 (front) **GND**	jumper 24-26 AWG	
S32		ESP32 (breakout) **GND**	jumper 24-26 AWG	
S32		MCP23017 U4 **VSS, A0, A1, A2**	jumper 24-26 AWG	address 0x20
S32		VL53L0X **GND**	jumper 24-26 AWG	
S33		Line sensor **GND**	jumper 24-26 AWG	
S35		Start button S2 **terminal 2**	jumper 24-26 AWG	
S36		Green LED D1 **- (short leg)**	jumper 24-26 AWG	
S36		Red LED D2 **- (short leg)**	jumper 24-26 AWG	
S37		Buzzer module **GND**	jumper 24-26 AWG	
S38		Bus-sense divider **bottom of 2.2k**	jumper 24-26 AWG	100 nF across the 2.2k

HX711 front-left

ID	Pin here	Goes to	Wire	Note
S16	VCC	3V3 bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17	GND	GND bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	SCK	ESP32 (breakout) **GPIO 5**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S19	DT	ESP32 (breakout) **GPIO 16**	jumper 24-26 AWG	
S23	E+	Load cell front-left **red**	the cell's own wires	
S23	E-	Load cell front-left **black**	the cell's own wires	
S23	A-	Load cell front-left **white**	the cell's own wires	
S23	A+	Load cell front-left **green**	the cell's own wires	

HX711 front-right

ID	Pin here	Goes to	Wire	Note
S16	VCC	3V3 bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17	GND	GND bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	SCK	ESP32 (breakout) **GPIO 5**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S20	DT	ESP32 (breakout) **GPIO 17**	jumper 24-26 AWG	
S23	E+	Load cell front-right **red**	the cell's own wires	
S23	E-	Load cell front-right **black**	the cell's own wires	
S23	A-	Load cell front-right **white**	the cell's own wires	
S23	A+	Load cell front-right **green**	the cell's own wires	

HX711 rear-left

ID	Pin here	Goes to	Wire	Note
S16	VCC	3V3 bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17	GND	GND bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	SCK	ESP32 (breakout) **GPIO 5**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S21	DT	ESP32 (breakout) **GPIO 18**	jumper 24-26 AWG	
S23	E+	Load cell rear-left **red**	the cell's own wires	
S23	E-	Load cell rear-left **black**	the cell's own wires	
S23	A-	Load cell rear-left **white**	the cell's own wires	
S23	A+	Load cell rear-left **green**	the cell's own wires	

HX711 rear-right

ID	Pin here	Goes to	Wire	Note
S16	VCC	3V3 bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S17	GND	GND bus	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S18	SCK	ESP32 (breakout) **GPIO 5**	jumper 24-26 AWG	bus on the tilting plate, 1 wire down
S22	DT	ESP32 (breakout) **GPIO 19**	jumper 24-26 AWG	
S23	E+	Load cell rear-right **red**	the cell's own wires	
S23	E-	Load cell rear-right **black**	the cell's own wires	
S23	A-	Load cell rear-right **white**	the cell's own wires	
S23	A+	Load cell rear-right **green**	the cell's own wires	

Load cell front-left

ID	Pin here	Goes to	Wire	Note
S23	red	HX711 front-left **E+**	the cell's own wires	
S23	black	HX711 front-left **E-**	the cell's own wires	
S23	white	HX711 front-left **A-**	the cell's own wires	
S23	green	HX711 front-left **A+**	the cell's own wires	

Load cell front-right

ID	Pin here	Goes to	Wire	Note
S23	red	HX711 front-right **E+**	the cell's own wires	
S23	black	HX711 front-right **E-**	the cell's own wires	
S23	white	HX711 front-right **A-**	the cell's own wires	
S23	green	HX711 front-right **A+**	the cell's own wires	

Load cell rear-left

ID	Pin here	Goes to	Wire	Note
S23	red	HX711 rear-left **E+**	the cell's own wires	
S23	black	HX711 rear-left **E-**	the cell's own wires	
S23	white	HX711 rear-left **A-**	the cell's own wires	
S23	green	HX711 rear-left **A+**	the cell's own wires	

Load cell rear-right

ID	Pin here	Goes to	Wire	Note
S23	red	HX711 rear-right **E+**	the cell's own wires	
S23	black	HX711 rear-right **E-**	the cell's own wires	
S23	white	HX711 rear-right **A-**	the cell's own wires	
S23	green	HX711 rear-right **A+**	the cell's own wires	

HC-SR04 (front)

ID	Pin here	Goes to	Wire	Note
S24	VCC	WAGO +5 V	jumper 24-26 AWG	5 V
S25	GND	GND bus	jumper 24-26 AWG	
S26	TRIG	ESP32 (breakout) **GPIO 4**	jumper 24-26 AWG	
S27	ECHO	ESP32 (breakout) **GPIO 23**	jumper 24-26 AWG	through a 1k / 2k divider (5 V -> 3.3 V)

VL53L0X

ID	Pin here	Goes to	Wire	Note
S30	SDA	ESP32 (breakout) **GPIO 21 (SDA)**	jumper 24-26 AWG	
S31	SCL	ESP32 (breakout) **GPIO 22 (SCL)**	jumper 24-26 AWG	
S32	VIN	3V3 bus	jumper 24-26 AWG	
S32	GND	GND bus	jumper 24-26 AWG	

Line sensor

ID	Pin here	Goes to	Wire	Note
S33	VCC	3V3 bus	jumper 24-26 AWG	
S33	GND	GND bus	jumper 24-26 AWG	
S34	OUT1	MCP23017 U4 **GPA0**	jumper 24-26 AWG	OUT1 = leftmost
S34	OUT2	MCP23017 U4 **GPA1**	jumper 24-26 AWG	
S34	OUT3	MCP23017 U4 **GPA2**	jumper 24-26 AWG	
S34	OUT4	MCP23017 U4 **GPA3**	jumper 24-26 AWG	
S34	OUT5	MCP23017 U4 **GPA4**	jumper 24-26 AWG	
S34	OUT6	MCP23017 U4 **GPA5**	jumper 24-26 AWG	
S34	OUT7	MCP23017 U4 **GPA6**	jumper 24-26 AWG	
S34	OUT8	MCP23017 U4 **GPA7**	jumper 24-26 AWG	OUT8 = rightmost

Start button S2

ID	Pin here	Goes to	Wire	Note
S35	terminal 1	MCP23017 U4 **GPB4**	jumper 24-26 AWG	internal pull-up
S35	terminal 2	GND bus	jumper 24-26 AWG	

Green LED D1

ID	Pin here	Goes to	Wire	Note
S36	+ (long leg)	MCP23017 U4 **GPB2**	jumper 24-26 AWG	through 330 ohm
S36	- (short leg)	GND bus	jumper 24-26 AWG	

Red LED D2

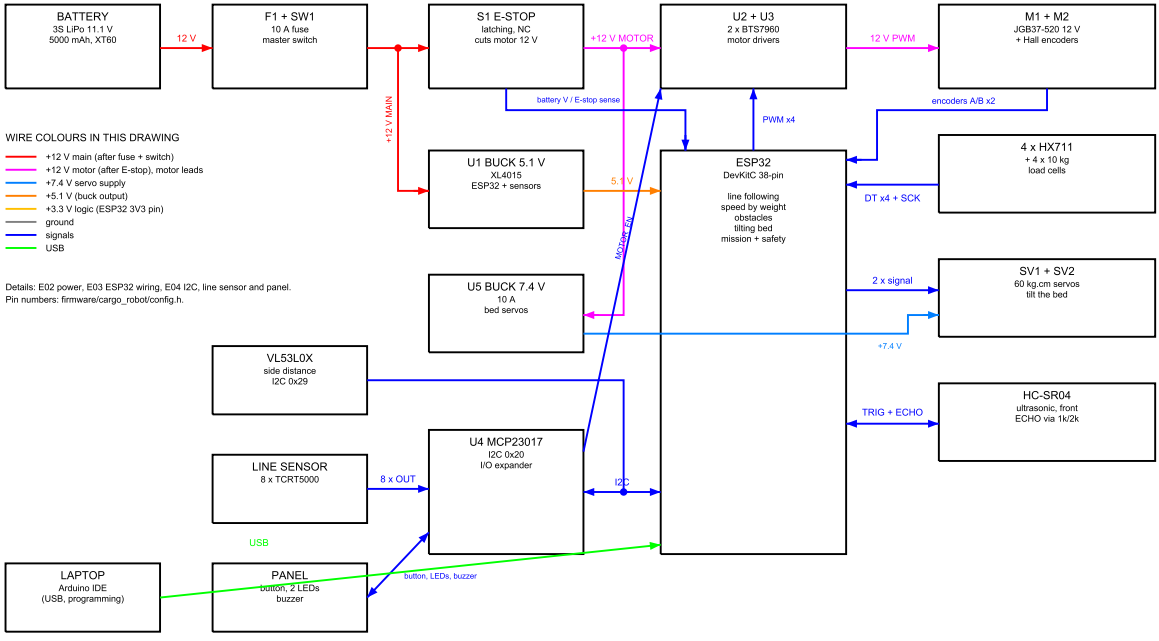
ID	Pin here	Goes to	Wire	Note
S36	+ (long leg)	MCP23017 U4 **GPB3**	jumper 24-26 AWG	through 330 ohm
S36	- (short leg)	GND bus	jumper 24-26 AWG	

Buzzer module

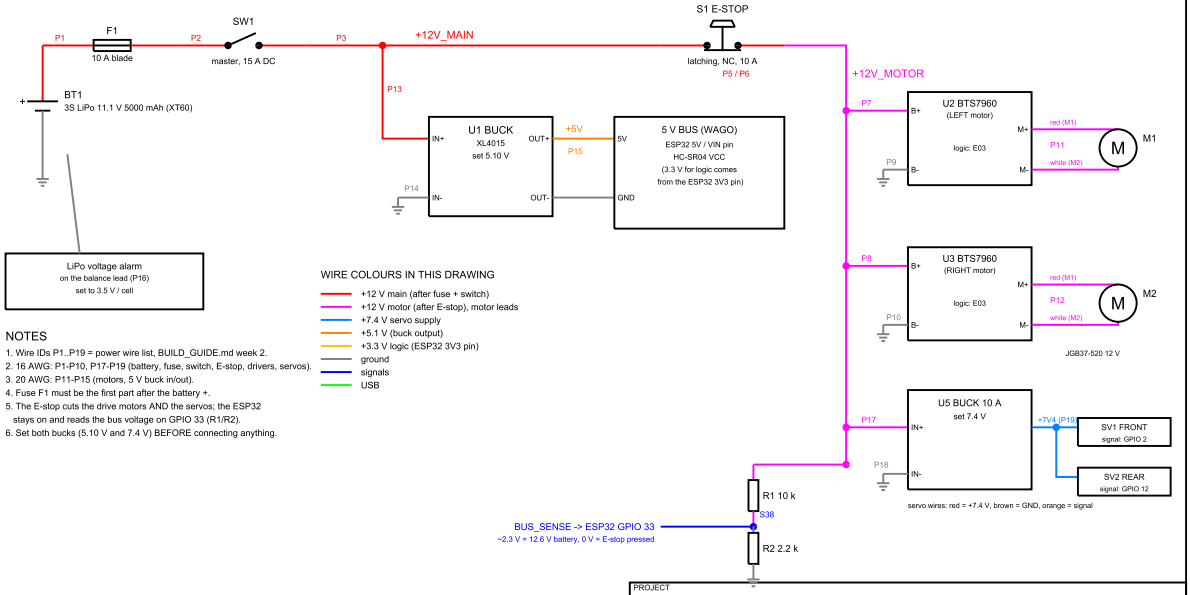
ID	Pin here	Goes to	Wire	Note
S37	VCC	3V3 bus	jumper 24-26 AWG	
S37	GND	GND bus	jumper 24-26 AWG	
S37	I/O	MCP23017 U4 **GPB1**	jumper 24-26 AWG	

Bus-sense divider

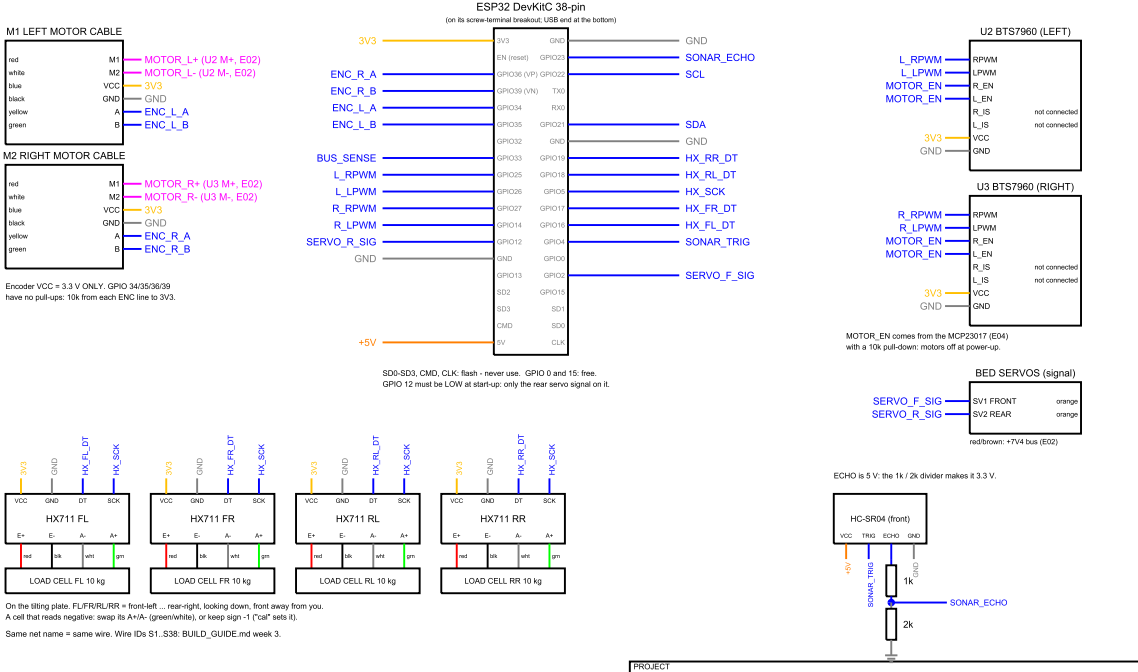
ID	Pin here	Goes to	Wire	Note
S38	top of 10k	WAGO +12 V MOTOR	jumper 24-26 AWG	after the E-stop
S38	bottom of 2.2k	GND bus	jumper 24-26 AWG	100 nF across the 2.2k
S38	junction	ESP32 (breakout) **GPIO 33**	jumper 24-26 AWG	about 2.3 V at 12.6 V; 0 V = E-stop pressed



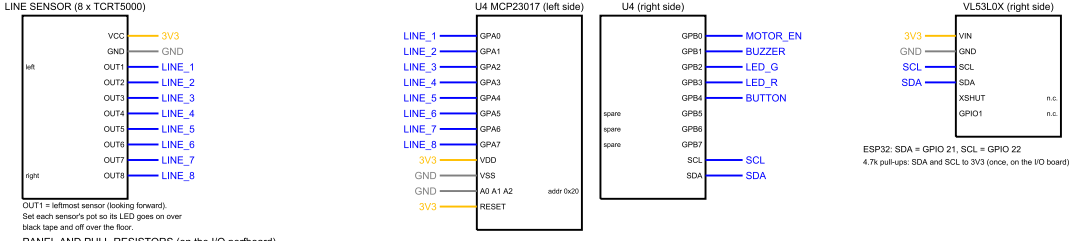
PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: SYSTEM BLOCK DIAGRAM					DWG No.: CR-V2-E01
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 1 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 22/10/2026	DRAWN:



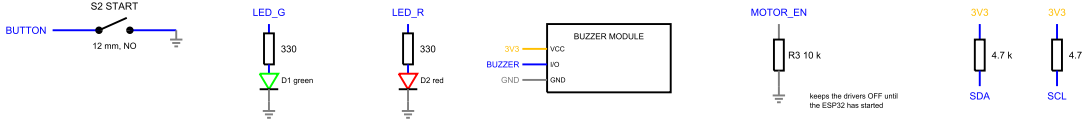
PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: POWER DISTRIBUTION					DWG No.: CR-V2-E02
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 2 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 22/10/2026	DRAWN:



PROJECT	AUTONOMOUS CARGO ROBOT - DESIGN v2				
TITLE	ESP32 WIRING				DWG No. CR-V2-E03
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	TYPE	DATE	DRAWN
NTS	3 / 4	-	SCHEMATIC	22/10/2026	



PANEL AND PULL-RESISTORS (on the I/O perfboard)



NOTES

1. All logic on this sheet runs on 3.3 V from the ESP32's 3V3 pin. Never feed 5 V to the MCP23017 or VL53L0X.
2. Line sensor: 1 = output HIGH over black tape (setting line_black). Check with the "line" command.
3. Same net name = same wire. Pin numbers: firmware/cargo_robot/config.h.

WIRE COLOURS IN THIS DRAWING

- +12 V main (after fuse + switch)
- +12 V motor (after E-stop), motor leads
- +7.4 V servo supply
- +5.1 V (buck output)
- +3.3 V logic (ESP32 3V3 pin)
- ground
- signals
- USB

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: I2C, LINE SENSOR AND PANEL					DWG No.: CR-V2-E04
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 4 / 4	UNITS: -	TYPE: SCHEMATIC	DATE: 22/10/2026	DRAWN: